CCST 102

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Section/Year: BSCS – 3B

Title

How AI Learns to Play Games

Chosen ML Application

The ML application I chose is Game-Playing AI. Because a AI learning how to play a game is a great way to showcase a Machine Learning Application. The AI or the computer can learn and decide what move/action to make in a game by looking at certain data, characteristic, information, state, or situation of the game and predict what will happen next. Some good examples are chess, minesweeper, and other games where each action, planning, and decision is important in the game.

Problem Description

Games are much harder to learn and predict because games are very complex and have a ton of choices or actions to do. Because a real person or player while playing, can look at the current state of the game where they can think, change plans, and adapt to the situation to win the game. If we think about coding a program to mimic a player, it is very hard and complex because we need a ton of rules and logic for the AI/Program to follow for every possible scenario or event. It is time consuming and hard. While if we used Machine learning, it will massively help the program/system to learn from looping events, learning from past events or examples, and improve over time by learning what moves/actions are the best one for the exact scenario.

Input and Output Identification

Input	Output
1. Current state of the game or board game	Best action, decision, move, or plan to execute
2. All possible or legal move/action	Predicting what the outcome or result from the possible actions
3. A players turn or action	Makes a decision/action from the players action
4. Location or positions of certain players or objects	A plan or decision around the object or interacting with the player
5. Past moves, actions, or history	Best decision to do in that scenario by learning from past mistake or history.

Traditional Programming vs. ML comparison

Traditional Programming: Input → Algorithm → Output

With traditional programming, the developer or coder writes the rules and logics that the system/program must follow and use. The developer must be in-depth and accurate with writing the rules so that the program will not make a mistake.

For an AI focused on playing a game, the programmer needs to write rules and logic around the game so that the program can decide on what decision or piece in a chess board to move depending on the game.

The input would be the status or state of the game's different elements, pieces, objects, players, etc. The algorithm is the set of rules and logics the program must follow. The output would be the result or decision made by the program.

Traditional programming is simple and works very well for easy tasks but falls off or becomes difficult to use when the task/situation is very complex and there are many choices, decisions, and actions to be made.

Machine Learning: Input + Output/Data → Machine Learning Algorithm

With machine learning, the program does not have a set algorithm or a set of rules and logics to follow. It instead learns from data, either from input or output data.

Now for the program, it needs input from the game, and output or data from a correct move from a chess match, loses or wins from a game, and even history of matches between players.

Machine learning algorithm focuses on studying or learning from a lot of data or examples so that it can learn the meaning, value, or relationship between the game and its actions.

Essentially, this makes the program/AI learn and improve over time without needing much rules, set algorithm, or too much human interference. It is self-sufficient and can be left alone to learn and improve.

Mini Dataset

Example from a Tic-Tac-Toe game:

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<tr><th>State of the Game</th><th>Available/Legal Moves</th><th>Best Move/Action</th><th>Result/Outcome</th></tr>
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Pattern/Relationship

The relationship or pattern in a Game-playing AI is that the input/action and what the current game state/situation are all connected to the best result, action, or move. The AI or program learns the pattern or relationship between these input/output data and the current game state by learning and studying it.

If a specific action or move leads to a win or victory, the program learns and memorizes that action/move that it will use it more often and strategize around it. This process shows how the AI/program recognizes patterns and relationship between the input/output and the environment, and adapt to them and make a decision fit for the situation.

Explanation of the Training Process

Input + Output → Training → Learning the Relationship/Pattern → Prediction

First, the machine learning algorithm learns from data and taking the input/output, such as the current game/board state or situation, and comparing through its past data/choices. Then after all that, the program chooses the best move to get close to the win/victory. During the training process, the model/program makes a prediction and then adjust or adapts when making mistakes and learning from the past.

As time passes, the model/program learns the relationship and patterns between the input and output. After the training process, the model/program now learned and can now use the inputs to predict the best and correct result/output much accurately or better.

Short Reflection

This activity helped me understand more about on developing a machine learning algorithm. That the Game-playing AI is a good way of learning and using machine learning because the AI learns from experience and data instead of using or depending on a set of rules and logics to follow.

It showed me how the AI can learn or study patterns/relationships, learn and improve over time, and makes better decisions/actions on different or difficult situations. While it is used specifically for games, this program/algorithm can actually work in real life and simulate fixing problems, such as planning or decision making in medical fields, transportation, education, etc.

Overall, machine learning is very important and useful for the real world or in real life.